

NENA Information Document for Location Validation Function Consistency

Abstract: This document provides recommendations that Location Validation Function (LVF) stakeholders, including operators, implementers, Geographic Information System (GIS) personnel and LVF clients can follow to help ensure that when LVFs from different vendors are provisioned with the same GIS data, they return consistent location validation responses for the same civic locations.



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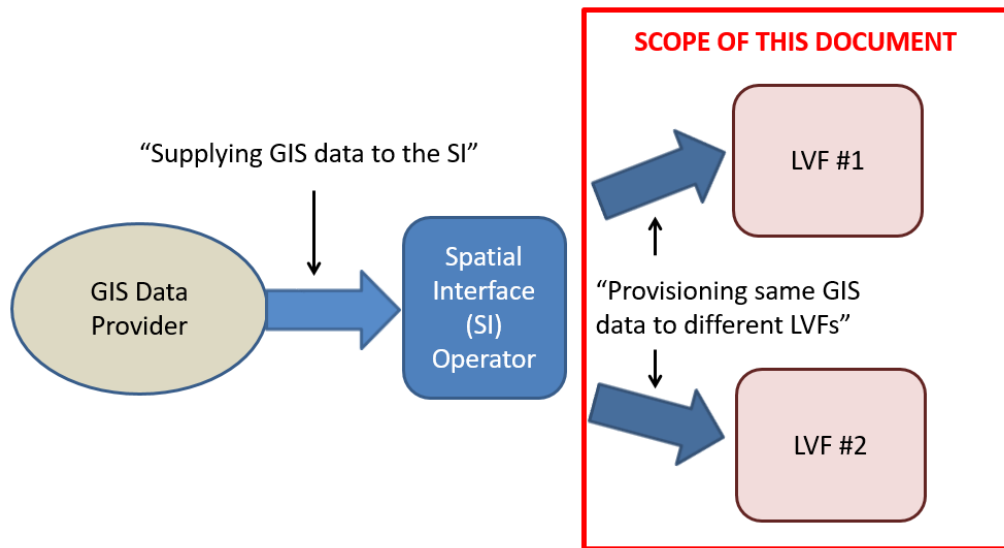
1 Executive Overview

This informational document builds upon the specifications provided in NENA-STA-010 [2] related to the Location Validation Function (LVF) and the Location to Service Translation (LoST) [7] protocol. As there is room for interpretation within these documents, the goal of this informational document is to outline recommendations that LVF stakeholders, including operators, implementers, Geographic Information System (GIS) personnel and LVF clients can follow to help ensure responses to LoST queries to different LVFs are consistent. It also provides recommendations for LVF client systems as to how the recommendations for the responses can be utilized.

This document, in addition to providing recommendations for consistency, also provides operational guidance to the stakeholders listed above. For example, this document advocates the adoption of draft-ietf-ecrit-similar-location [10]. Adoption of this LoST extension is viewed as critical to the LVF's ability to effectively validate addresses for Next Generation 9-1-1 (NG9-1-1) calls.

Purpose and Scope

The premise of this informational document is to provide guidance to implementers of LVF systems, such that when LVFs from different vendors are provisioned with the same GIS data, they return consistent responses for the same civic location. However, this informational document does not address differences in how GIS data from a GIS Data Provider can be supplied to a Spatial Interface (SI), nor does it address differences in how a SI Operator can process data in a SI. Instead it assumes that the data coming out of the SI (i.e., provisioned to the LVF) does not change and is the same. Changes to either of those processes can change the data being provisioned to the LVF, thereby indirectly impacting location validation consistency.



Benefits

Consistency between the LVFs implemented by various vendors is of great benefit to all parties involved in developing and maintaining systems and data for use in NG9-1-1. While many of these benefits may not be immediately visible, they become especially evident when switching between LVF providers. While a change in LVF providers may require revalidation, this revalidation should not result in different validation results when provisioned with the same GIS data through the same SI feed. Additionally, there may be more than one LVF provider that services a network. Such variability would become a significant problem for 9-1-1 Authorities, GIS Data Providers, SI Operators, Location Information Server (LIS) Operators and LVF Operators. It would introduce not only location validation inconsistency, but it would also delay such a change in LVF Operators, increase the amount of time and effort to clean up LIS and GIS databases that were previously considered valid, and it would decrease the confidence by all parties in the validation process. This document has been developed in order to minimize and hopefully eliminate these issues.

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Reason for Issue/Reissue

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2 Location Validation Function Consistency

2.1 Why Validate?

The concept of validating the locations of potential 9-1-1 callers, before calls are ever placed, began with the inception of Enhanced 9-1-1 (E9-1-1). E9-1-1 introduced the notion of the Master Street Address Guide (MSAG) which is a reference address format specific for 9-1-1 purposes. All carriers were required to make their customers' civic address to match an entry in the MSAG so that the proper Emergency Service Number (ESN) could be assigned to the customers' telephone numbers. By doing so, a 9-1-1 call made from a Telephone Number (TN) associated with a MSAG-validated address would be routed to the appropriate or designated PSAP and properly displayed to the call taker through the Automatic Location Identification (ALI) in a standardized format.

In NG9-1-1, the Location Information Server (LIS) (or the location server/database component of a Legacy Network Gateway) may provide the subscriber location information to the PSAP telecommunicator instead of an ALI server. The same location provided by the LIS is used by the NG9-1-1 Core Services (NGCS) to route the call to the appropriate or designated PSAP. Alternatively, the device may convey its own location directly. The Location Validation Function provides validation service in an NG9-1-1 system and is the logical equivalent of the ALI using the MSAG to validate. We say that a location must be LVF valid before being placed in a LIS, in the same way we say that a location must be MSAG valid before being placed in an ALI.

In NG9-1-1, the data structures for describing location are different from those used in E9-1-1, and what was a description of the ALI location format now is represented by a "PIDF-LO", which is a data structure used for carrying both civic (street address) and geo (lat/lon/altitude) forms of location. PIDF-LO, in turn, conforms to element use defined by CLDXF, which is, among other things a profile of PIDF-LO for use for US addresses. A Canadian version of CLDXF will be created. CLDXF, in turn, is a profile of the Federal Geographic Data Committee (FGDC) standards for US Addresses. As a result, a location that is MSAG valid is typically not LVF valid and vice versa.

Validating the correctness of locations stored within a location database is of vital importance. In both E9-1-1 and NG9-1-1, invalid locations associated with subscriber records can significantly impact emergency response, making it time-consuming or impossible to dispatch first responders to the locations of calls, and in NG9-1-1, invalid locations can result in call routing failures and complicate the formatting of call taking displays. For these reasons, a 9-1-1 Authority requires validation of carrier subscriber records before they are placed in ALI or LIS servers or other systems supplying the location data via an ALI or LIS.

In NG9-1-1, other stakeholders may wish to validate locations prior to, or during, the handling of a call for service, such as GIS Authorities or telecommunicators themselves. These stakeholders and their roles are identified in section 2.3.1.

2.2 What is a valid location?

The notion of LVF validity is a complex subject, which is confusing to those new to NG9-1-1. LVF valid is a technical state where a query to an LVF using the LoST protocol returns no components in the <invalid> list of the <locationValidation> element of the <findServiceResponse>. If there are no elements in <invalid>, the location is considered "LVF-Valid". The LVF, in turn, uses its source data (provisioned via the SI) to determine the element returned in the <invalid> list.

LVF validity is primarily used by a location provider, the operator of a LIS. If a LoST query returns no element in <invalid>, the process of determining if a location should be added to the LIS is complete – it's valid and can be added to the LIS and used on a subsequent emergency call. This operation may also be used by a Telecommunicator when entering an address manually. If the location returns no element in <invalid> from a LoST query, the Telecommunicator can stop – there is sufficient information to get an LVF valid location.

Since the LVF determines the validity of a location by comparing the input location data with the data that is provisioned into it, it is up to the 9-1-1 Authority to determine what the LVF should consider valid via implementation by the GIS Data Provider. The data entered in the GIS determines what the LVF considers to be valid.

An important consideration is that a LoST query can, and will, return a route even if the location is invalid. The ECRF (and the LVF, since it does return a route) will take whatever information is provided in the query to determine the best route it can provide. Invalid locations get a route. So, the requirements on the data for validation purposes can be, and often are more stringent than the data supplied for a route. To get an LVF valid location, there has to be enough information to dispatch a responder and get them to the right place. Note that the telecommunicator can choose to dispatch a responder to an invalid location if wanted.

Consider a GIS that only has road centerlines. The LVF provisioned via the SI from that GIS will consider an address with the correct country, state, county, municipality, road name (with appropriate pre and post directionals and prefix/suffix) to be valid so long as the address number falls within a range of a centerline record for that road. Even a non-existent address number within the range will return no entries in <invalid>.

Now consider a GIS system that has both road centerlines and a complete set of address points for a segment of road. If the LIS queries the LVF with an address number not matching an address point, though falling within the road centerline range, then the result

is valid. Even though no address point matched, the LVF found a road centerline segment that did match and thus the address is valid.

If only address points are in the GIS, and road centerlines are not, then the address will only be valid if there is an address point that matches the address number in the query. Remember that the LVF will return a route even if the address number doesn't match an address point. It would return the address number in the <invalid> list. That means the LIS operator should not enter the address in the LIS. It should verify the address number and file a discrepancy report on the LVF if it believes the address is correct. However, if the address with a bad address number was used in a 9-1-1 call, the ECRF would provide a route for that call, if it has enough information to do so.

This same logic can be carried to "subaddress" elements. If there is an address point with just the address number, and a set of address points for each of the buildings at that address number, then an address with no building number or a non-existent building number will be considered valid. If that's what the 9-1-1 Authority wants, then the data is fine. If the 9-1-1 Authority wants the LIS operator to always enter a valid building number, for a particular building, then it should remove the address point without a building number as well as the associated road centerline and force the <invalid> list to have "Building" in it if a query without Building is presented to the LVF. The same logic can be applied to floors, unit numbers and other subaddress elements.

It's up to the 9-1-1 Authority to decide what data the LIS operator validates against in order to put a record in the LIS. The GIS Data Provider provisions to the LVF via the SI. The LVF interprets the data to decide how to populate <invalid>. The LIS operator uses the results of the LoST response to decide what is good enough to put in the LIS, and thus what will appear when a 9-1-1 call is made from that location.

2.3 Operational Considerations

2.3.1 Stakeholders and Roles

2.3.1.1 Origination Networks and Service Providers

NENA-STA-010 [2] prescribes that all Origination Networks and Service Providers are required to validate and periodically revalidate all static VoIP and wireline subscriber location records prior to being loaded into the databases serving up the locations. Nomadic VoIP locations should also be validated anytime the calling device's location has changed. Validation of the new location would be accomplished by the LIS operator using the LVF. Wireless carriers must validate the location of cell towers, provisioned for Phase I location delivery via a pseudo-ANI (pANI), Emergency Service Routing Key (ESRK) or Emergency Service Routing Digits (ESRD) provisioning.

2.3.1.2 NG9-1-1 Functional Element Operator

An NG9-1-1 Functional Element Operator is an entity providing one or more of the following 9-1-1 elements: network, NGCS, call handling, incident handling or any other functional elements within a PSAP, or a database service. It is common for Service Providers as well as 9-1-1 Authorities to hire NG9-1-1 Functional Element Operators to do various tasks on their behalf.

2.3.1.2.1 LIS Operator

A LIS Operator operates the LIS associated with the origination network used by the subscribers attached to it. Per NENA-STA-010 [2], the LIS is the responsibility of the Origination Network or Access Service Provider whose subscribers must have associated locations, which for the purposes of this document, would be validated using the LVF before being provisioned to their LIS. LIS records must also be periodically revalidated to ensure any changes to the GIS data provisioned to the LVF are incorporated in their subscriber's location records¹. Note that the location server/database component of an LNG is equivalent to a LIS when it contains civic location data, and the LNG operator is then equivalent to a LIS Operator.

2.3.1.2.2 LVF Operator

An LVF Operator provides the LoST protocol server to be utilized by LIS Operators for validating subscriber locations. The LVF may be operated by various entities, such as a 9-1-1 authority, a NG9-1-1 Core Services Provider, or a third party.

9-1-1 Authority

A 9-1-1 Authority is a State, County, Regional, Tribal or other governmental entity responsible for 9-1-1 service operations in a given area. The 9-1-1 Authority may assume the role of GIS Data Provider for the service area boundary, or it may delegate that function to another entity. Similarly, for civic location data, the 9-1-1 Authority may work with all the authoritative data sources within its service area to create and maintain civic location layers, or it may designate one or more GIS Data Providers and specify the provisioning area for which each is responsible.

In situations where there are multiple GIS Data Providers, for example where local data are aggregated to a regional or statewide service area, the 9-1-1 Authority needs to specify exactly how and by whom local data will be reviewed, edge-matching or other boundary issues resolved, and final quality control performed before the data are provided to the SI. For example, the 9-1-1 Authority may determine that data must be returned to the original GIS Data Provider for review and correction if edits are needed, or the 9-1-1 Authority may assign the responsibility for reviewing data and making corrections over the

¹ The IETF has developed a mechanism allowing a LIS to be notified when records will be affected by a change in GIS data. Please refer to section 2.3.5.2 for more details.

entire service area to one of several GIS Data Providers, or it may assign that role to some other entity.

GIS Data Provider

A GIS Data Provider is an entity that creates and maintains GIS data, or that aggregates and compiles data provided by others based on authoritative sources, into a NENA standards-compliant format to be provided to the SI. If there are multiple candidate sources of civic location and address data within the same area, the GIS Data Provider must determine which ones are authoritative. Whether or not the GIS Data Provider is itself the authoritative source for a given dataset, the GIS Data Provider is responsible for maintaining consistency between the authoritative data sources and the GIS data provided to the SI. A GIS Data Provider may provide civic location data for all or only a portion of the service area. Each GIS Data Provider is responsible for performing quality control checks on the data within their area and ensuring that it conforms to NENA standards. Data from a GIS Data Provider must be delivered in final form, ready for use within the ECRF and LVF.

2.3.1.3 Telecommunicator

A telecommunicator is a person employed by a PSAP and/or a response agency qualified to answer incoming emergency calls and/or provides for the appropriate emergency response.

A telecommunicator may elect to validate civic locations while handling a call for service or incident. For example, a wireless caller may provide her/his location in the form of a civic address via audio or text. The location provided by the caller, in this case, will not have been obtained from a LIS and relayed to the PSAP upon receiving the call. Since the location will instead be manually recorded and entered within a call or incident handling function, the telecommunicator (or the software on the telecommunicator's behalf) may elect to verify its correctness by using an LVF.

2.3.2 NG9-1-1 Location Validation

Per NENA-STA-010 [2], the sole purpose of the LVF is to facilitate the process of NG9-1-1 location validation. A LIS or other Database Management System (DBMS) validates the correctness of location records by using an LVF to determine if the locations are valid. Valid records may be inserted into the LIS or location database, while invalid records must be corrected. A LIS validates (and periodically revalidates its records) using the LoST protocol as defined in RFC 5222 [7].

The end result is no different than using the MSAG for Service Order Input validation prior to records being provisioned to a legacy ALI database. Simply, different mechanisms and databases are utilized to do the validations in NG9-1-1.

Also note that validation is only provided for civic locations in the LVF. The LVF as currently specified does not support validating latitude/longitude or other geodetic locations.

2.3.3 Corresponding Legacy Validation

Unlike NG9-1-1, legacy E9-1-1 location validation is associated to the MSAG and use of ALI and VoIP Positioning Center (VPC) databases. An Origination Networks and Service Provider or their designated 9-1-1 service provider validates subscriber addresses using the MSAG. Subscriber records and pANIs should only be provisioned to an ALI or VPC after passing MSAG validation and only provisioned using the exact structure as provided in the MSAG.

2.3.4 NG9-1-1 LVF Validation Considerations

NG9-1-1 changes the way that locations are validated. Because the mechanism differs from the one used in legacy E9-1-1, it is important to consider a few key points. The following operational considerations are inherent given the design of the i3 architecture and the transition to it, and stakeholders must be mindful of them as they transition to NG9-1-1 and LVF-based location validation.

- The software that performs the job of validating locations must validate them by interacting with an LVF instead of an MSAG. In the legacy environment, the Origination Network and Service Provider sends their subscriber records to the System Service Provider (SSP) to be placed in the regional ALI. In NG9-1-1, the LVF validated records are the responsibility of the Origination Network and Service Provider. These records are placed in one or more LIS, which cover their entire service footprint, be it large or small. The software used by Origination Networks and Service Providers or 9-1-1 Service Providers must be capable of issuing LoST requests to an LVF, interpreting its response and support making changes to the location record as required.
- The LoST protocol does not support batch location validation; rather, locations are validated one at a time. This differs from E9-1-1 batch processing via Service Order Input (SOI). The software used to validate locations will need to account for this or operational procedures may require modification. This also permits location validation at order entry time, or even in a self-service ordering system.
- The LVF utilizes GIS data to determine if locations are valid. Validation data is derived from the information in the GIS system. In NG9-1-1, this means that GIS Data Providers, along with local GIS Authorities, become stakeholders within the location validation process, and must be considered when resolving location validation issues or other discrepancies. Historically, the MSAG coordinator has fulfilled the role of maintaining validation data. Depending on the local jurisdiction, this role may persist. If so, it will require a coordinated relationship with the Local

9-1-1 Authority and/or the GIS Data Provider due to the transition from the tabular MSAG to the GIS data for validation.

- When GIS data changes, and the changes are provisioned to the LVF, previously-valid locations may become invalid. The IETF is working on a methodology described in Validation of Locations Around a Planned Change [11] for notifying a LIS Operator of a change to GIS data that would trigger a new validation request. Otherwise, the LIS Operator will be responsible for periodically revalidating location records in the LIS.
- Discrepancy reporting should follow the guidelines set forth in NENA Next Generation 9-1-1 Data Management Requirements, NENA-REQ-002 [12] and mechanisms described in NENA-STA-010 [2]. Resolving discrepancies may often require coordination between various stakeholders and may be subject to a specific set of recommended turnaround times, as also defined in NENA-REQ-002 [12].
- An Origination Network and Service Provider may provide service over a geographic region large enough that it must interact with multiple LVFs when validating locations. As specified in RFC 5222 [7], a single geographic region may be covered by one, and only one, authoritative LVF – but the Origination Networks and Service Provider’s region of coverage may encompass a larger area. In the i3 end state, a National Forest Guide facilitates the discovery of the LVF responsible for a particular geographic region. Until such time that a National Forest Guide is established and is available for use, additional configuration management of LVF clients or LVFs themselves may be required. In the absence of the National Forest Guide, software that performs location validation must be provisioned with one or more LVFs to interact with, in the form of one or more U-NAPTR, as specified in RFC 5222 [7], application unique strings. The application unique strings of multiple LVFs may be required if the LIS Operator, or other LVF client validates locations in different geographic regions. Alternatively, the software that validates locations could interact with a single LVF that is itself provisioned with the coverage regions of other authoritative regions, such that it is able to recurse or redirect to the appropriate authoritative LVF when responding to a location validation request. Clients should never start at the Forest Guide. They should query what they believe to be the correct LVF, or an LVF who has agreed to handle “out of area” validations for them and allow that LVF to query the Forest Guide if it needs to.
- It is important to remember that validation is not routing. Locations that are considered invalid may route correctly, so long as there is sufficient information in the ECRF to determine where the caller is. Postal locations, MSAG locations and other data may very well route correctly. However, such locations should never be loaded in a LIS. Only a true LVF valid location should be loaded in the LIS, and the system is designed to be precise and definitive with respect to validation. Validation

happens before the call, when there is time to make sure the address is specified correctly and represents the right location.

In addition to the previous considerations, it is important to ensure that location validation remains consistent across LVF implementations and deployments, both during the transition to NG9-1-1 and in the end state. Gaps exist within published standards such as RFC 5222 [7] and NENA-STA-010 [2] that leave aspects of location validation – how an LVF performs it, and how an LVF is configured – open to interpretation. Without further guidance, an LVF implemented by one vendor, or managed by one authority, may produce location validation results that are inconsistent with an LVF implemented by a different vendor, or managed by a different authority. The operator of software that validates locations against one or more LVFs may experience such inconsistencies if:

- The LVF that the operator utilizes is replaced by an LVF provided by a different vendor or operator.
- The operator is responsible for multiple geographic regions and through the course of validation locations, interact with multiple LVFs, either provided by different vendors or operated by different agencies.

These inconsistencies may lead to a sudden and unexpected increase in the number of invalid location records in the LIS, if switching from one LVF vendor / operator to another, resulting in additional work to correct them (potentially involving multiple stakeholders). Inconsistencies experienced on a day-to-day basis, if interacting with multiple LVFs, may lead to software interoperability issues or general, unnecessary effort on the part of staff tasked with validating records to interpret differences in results.

The purpose of this document is to further describe this important consideration and offer recommendations for mitigating it, to ensure a successful transition to NG9-1-1 and LVF-based location validation.

2.3.5 Possible Future Enhancements to the LoST Protocol

Since originally published, RFC 4119 [4] has been extended several times by additional IETF work including RFC 5139 [5] and RFC 5491 [6]. While similar extensions to the LoST protocol defined in RFC 5222 [7] have not yet reached RFC status, it will be important for readers of this document to remain aware of relevant developments from the IETF. This will be especially critical for potential extensions to LoST and their impacts on and benefits for location validation. This includes items in section 3.4 of this document describing recommendations for additional development work. Below are references to two draft documents that would have a significant impact on location validation should they reach RFC status.

2.3.5.1A LoST extension to return complete and similar location info [10]

Abstract: "This draft proposes a new way to provide returned location information in LoST responses that is either of a completed or similar form to the original input civic location, based on whether valid or invalid civic location elements are returned within the findServiceResponse message. This document defines a new extension to the findServiceResponse message within the LoST protocol [RFC5222] that enables the LoST protocol to return a completed civic location element set for a valid location response, and one or more suggested sets of similar location information for invalid LoST responses. These two types of civic locations are referred to as either "complete location" or "similar location", and are included as compilation of ca type xml elements within the existing LoST findServiceResponse message structure."

The use of this extension is to

- a) return any missing elements of a location that has sufficient information to match a single address within the LVF. The client should use the returned location information to populate the LIS
- b) suggest possible valid locations that may be what the user of the client (the human trying to enter a location in the LIS) meant when a location is invalid.

2.3.5.2 Validation of Locations Around a Planned Change [11]

Abstract: "This draft proposes an extension to LoST (RFC5222) that allows a planned change to the data in the LoST server to occur. Records that previously were valid will become invalid at a date in the future, and new locations will become valid after the date. The extension adds two elements to the <findService> request: a URI to be used to inform the LIS that previously valid locations will be invalid after the planned change date, and add a date which requests the server to perform validation as of the date specified. It also adds an optional TTL element to the response, which informs all queriers the current expected lifetime of the validation."

The use of this extension is to provide a notification to a LIS that a location previously validated may no longer be valid, or could become invalid in the future. It provides an alternative to periodic revalidation.

2.4 Current Validation Consistency Challenges

As described in section 2.3.4, the lack of additional guidance regarding location validation may result in inconsistent results produced by LVFs constructed by different vendors or managed by different operators. The results of LVF validation may vary in two basic ways: baseline validity and in the contents of the valid/invalid/unchecked element lists found in an LVF's LoST findServiceResponse. Baseline validity, in this document, describes the simple notion of whether or not a civic location is valid. Either the location is valid, or it is not. LVFs may produce different baseline validation results if one LVF claims a civic

location is valid, while another LVF simply claims it is invalid. Consistency of the contents of the valid/invalid/unchecked element lists is more granular. Two LVFs may claim that a civic location is invalid, but they may claim that different civic location elements are valid, invalid, or unchecked.

2.4.1 Causes of Inconsistent Baseline Validation Results

The following examples illustrate situations in which an LVF may claim an address is valid, while another LVF claims the address is invalid, when both LVFs are provisioned with the same GIS data.

2.4.1.1 Lack of Minimum Element Requirements

No document or standard currently requires a PIDF-LO civic location to contain a minimum set of elements before an LVF will attempt validation, but unofficial recommendations have been adopted by various LVF implementations. Some LVF implementations may require a minimum set of information to be present in a civic location, and will claim it is invalid if such information is omitted. Other LVF implementations may not impose such constraints. For an example, see Appendix A section 1A.1.

Recommendation: LVF vendors shall follow the guidelines in section 2.5.

2.4.1.2 Use of Administrative Boundaries to Determine Validity

No specific guidance exists that describes how an LVF should utilize administrative boundaries when validating locations. One LVF implementation may simply examine address point and road centerline attributions when validating addresses. Another LVF implementation may examine those attributes, and also compare the location of the matched GIS feature with the administrative boundary features it intersects, to ensure that all three elements – the address, the GIS feature, and the administrative boundaries – all share the same administrative elements. For an example, see Appendix A section 1A.2.

Recommendation: LVF operators shall utilize only record attributes and ignore geometry when LVF validating a civic location.

2.4.1.3 Variance in Geocoding Offsets

An LVF's response to a validation query contains one or more service mappings, which are a key component of the LoST protocol defined in RFC 5222 [7]. Service mappings describe the relationship of locations and service URNs to service boundaries and URIs. When an LVF matches a civic location against a road centerline, the process of finding a service mapping to return in the response may involve "geocoding" the address – determining a geodetic (X/Y) position based upon house number's position within the address ranges of the matching road centerline. Many geocoding algorithms offset the geodetic position a certain distance to either side of the road centerline (depending on the parity of the address), and a certain distance from the ends of the road centerline. LVF

implementations might allow configuration of the offset values, and differences in the configured values could lead to inconsistent baseline validation results. Small geocoding offsets (whether side offsets or end offsets, also known as “squeezing”) might always return interpolated points that intersect service boundaries, allowing an LVF to return one or more service mappings in its validation responses. Large geocoding offsets, however, may result in interpolated points that fall outside of an LVF’s authoritative coverage region or outside of its service boundaries. In that case, an LVF may not be able to return a service mapping, and instead may return a LoST notFound error. LoST errors cannot contain location validation results, and for this reason, variance in geocoding offsets may prevent some validation queries from being successfully answered. For an example, see Appendix A section 1A.3.

Recommendation: GIS Data Providers should ensure that site/structure address points exist for addresses along road centerlines that are coincident with, or near, the edges of service boundaries. The road centerlines should be attributed to specify that validating addresses along them require site/structure address point matches, per the recommendation of adding a road centerline “validation” attribute to NENA-STA-006 [9] in section 3.4 of this document. If site/structure address points are not created along these road centerlines, and the road centerline “validation” attribute is not utilized, LVF results may be inconsistent based upon variance in the ways that LVFs determine which service mapping(s) to return in a LoST findServiceResponse. This document also recommends, in section 3.4, that additional work be taken to recommend or standardize the approaches used to determine the service mapping(s) to return in validation queries. Until such time that further guidance exists, LVF Providers that operate LVFs that geocode civic locations against road centerlines to determine the service mapping(s) to return should ensure that geocoding offsets are never large enough that a geocoded point falls outside of an LVF’s service area or a service boundary within its coverage region. SI Operators, if possible, should attempt to discover instances where address points are not available and large geocoding offsets could cause location validation problems, especially along boundaries. The SI Operators should recommend to GIS Data Providers that these situations be remediated by creating site/structure address points and using the recommended road centerline “validation” attribute.

2.4.1.4 Use of Alias Names to Determine Validity

No current guidance exists that suggests street name aliases should, or should not, be examined when validating a civic location. This means that LVF implementations may vary; some may elect to validate against primary names only, while other LVFs may validate against street name aliases as well. For an example, see Appendix A section 1A.4.

Recommendation: Aliases shall not be considered when determining if a proffered value matches the GIS data, during location validation. That said, it is recommended that aliases be used in conjunction with the “Similar Locations” IETF document [10] to determine

potential matches to be returned along with an invalid LoST response. As noted in section 2.5.9, an LVF returning similar locations shall not return addresses containing alias names – rather, only valid addresses shall be returned. It is also recommended that an ECRF use alias names at call routing time, to determine the best route.

2.4.1.5 Differences in GIS Field Mappings

In the normal course of initially setting up and administering an LVF, a system administrator will configure the attribute fields to examine within its GIS data layers when searching for a particular civic location value. For example, an administrator might specify a field mapping between the A3 element and the incorporated municipality attributes of the address point and road centerline layers, or the administrator may choose to map the A3 element to the MSAG Community legacy attribute fields. Differences in configuration between one LVF and another could cause inconsistent baseline validation results between the LVFs. For an example, see Appendix A section 1A.5.

Recommendation: Per section 2.5.5, an LVF's civic location elements should be mapped to the CLDXF-compliant [8] road centerline and site/structure address point attributes.

2.4.2 Examples of Inconsistent Valid / Invalid / Unchecked Elements

The following examples illustrate situations in which LVFs may claim an address is either valid or invalid (in that, they return the same baseline validation result), but they claim that different address elements are valid, invalid, or unchecked. All LVFs in the examples are provisioned with the same GIS data.

2.4.2.1 Reporting Address Elements in a Different Order

No document or standard currently requires or recommends that LVFs report an address' elements in a consistent order, such as in a top-down hierarchy from coarse-grained administrative areas to fine-grained subaddress elements. However, unofficial suggestions have been adopted by various LVF implementations that examine elements using such a hierarchy, and report the elements that are valid, invalid, and unchecked in a consistent order. Without further guidance, it is likely that different LVF implementations will return the same baseline validation responses, but each will report the elements that are valid, invalid, and unchecked in a different order. For an example, see Appendix A section 1A.6.

Recommendation: To ensure that LVFs return the same elements in the valid, invalid, and unchecked lists, in the same order within each list, LVFs shall validate addresses using a hierarchical approach, as outlined in section 2.5.3.

2.4.2.2 Claiming Different Elements are Invalid

A civic location may contain multiple valid and invalid elements. While NENA-STA-010 [2] section 4.4.4 describes valid, invalid, and unchecked elements, it does not prescribe that all valid elements must be returned by an LVF, and that all (or only one, or a subset) of invalid

elements must be returned by an LVF. For the purposes of this example, no document or standard currently requires or recommends which of the invalid elements should be returned; one LVF implementation may claim that only a single element is invalid and the rest as either valid or unchecked, while another LVF implementation might report more than one is invalid. For an example, see Appendix A section 1A.7.

Recommendation: It is recommended that the LVF follow the contextual hierarchy of elements defined in section 2.5.3, and if the LVF encounters an invalid civic location element during validation, it shall stop any further validation of the address. As such, all elements examined before discovering the invalid element would be considered valid, only the first encountered invalid element would be considered invalid, and all remaining elements, if any, would be considered unchecked. This recommendation mitigates the potential complexity of determining the list of invalid elements to return and the ambiguity that may come with determining which elements are invalid within a particular valid context. Since it is possible that multiple location records in the LVF could be drawn upon to determine what would be populated in the invalid list, it would be difficult, at best, to create recommendations about how to determine which record to leverage to populate the invalid list, and in any case, differences in LVF design and implementation would lead to potentially-inconsistent validation responses. Limiting the response to having one and only one invalid element is one way to help ensure that LVFs return consistent responses.

2.5 Location Validation Guidelines

When an LVF validates a civic location, the following general guidelines apply:

2.5.1 Use of Alternate Data Sets to Determine Validity

Use of data sets other than the authoritative GIS data in LVF validation is discouraged. The use of alternate sources, such as MSAG or postal data to perform LVF validation, whether primary or backup to the authoritative GIS data, introduces the potential for valid and/or different responses that could vary from a response that only leverages the authoritative GIS data from the 9-1-1 authority.

2.5.2 Elements Must Match One-to-One

A valid civic location has address element values in PIDF-LO [4] which match, ignoring case, the same values in one and only one record within the LVF, within either the site/structure address point layer or road centerline layer. All civic location elements in a proffered civic location must match those in the LVF record; if only a subset match, the location is not considered valid, except when elements exist in the proffered locations that do not have corresponding attributes in the matching LVF record. For example, if a proffered civic location includes a house number suffix, and the LVF matches the address on one and only one road centerline feature, the LVF will consider the address valid but claim the house number suffix is unchecked, because road centerline features lack house

number suffix attributes. If this is not the behavior desired by the 9-1-1 Authority, address points should be used and the corresponding road centerline should be removed. See similar comments in section 2.2.

Note: When locations match road centerlines instead of site/structure addresses, the LVF will not check the validity of the following elements. A value of unchecked will be provided for the following CLXDF-compliant [8] PIDF-LO [4] civic location elements (provided in PIDF-LO element order):

PIDF-LO Element Name	CLDXF Element Name
HNS	Address Number Suffix
LMK	Complete Landmark Name
LMKP	Landmark Name Part
LOC	Additional Location Information
FLR	Floor
BLD	Building
UNIT	Unit
ROOM	Room
SEAT	Seat
PLC	Place Type
MP	Milepost

2.5.3 Use the Elements' Contextual Hierarchy to Report Results Consistently

Civic location elements have an implicit hierarchy, from coarse-grained to fine-grained. The hierarchy starts at the top with coarse-grained elements like Country, followed by State, and continues to finer-grained elements like Street Name, down to subaddress elements like Room, Seat, and Additional Location Information, followed at last by miscellaneous elements like Place Type. When reporting the results of a validation query, an LVF adheres to the hierarchy and report any valid, invalid, and unchecked elements in hierarchical order, so that coarse-grained elements are returned before fine-grained elements. This will help to ensure that all LVF implementations return the same results. The contextual hierarchy of elements to validate in PIDF-LO [4] as described in CLDXF [8] is:

PIDF-LO Element Name	CLDXF Element Name
country	Country

PIDF-LO Element Name	CLDXF Element Name
A1	State
A2	County
A3	Incorporated Municipality
A4	Unincorporated Community
A5	Neighborhood Community
RD	Street Name
PRM	Street Name Pre Modifier
PRD	Street Name Pre Directional
STP	Street Name Pre Type
STPS	Street Name Pre Type Separator
STS	Street Name Post Type
POD	Street Name Post Directional
POM	Street Name Post Modifier
HNO	House Number
HNP	Address Number Prefix
HNS	Address Number Suffix
MP	Milepost
BLD	Building
LOC	Additional Location Information
FLR	Floor
UNIT	Unit
ROOM	Room
SEAT	Seat
PCN	Postal Community Name
PC	Postal Code
LMK	Complete Landmark Name

PIDF-LO Element Name	CLDXF Element Name
LMKP	Landmark Name Part
PLC	Place Type

In addition, the NENA NG9-1-1 CLDXF standard restricts the IETF PIDF-LO civic address standard by exclusion of the following PIDF-LO elements:

- A6 – Street (group of streets below the neighborhood level): Not used in the United States
- ADDCODE – Address Code: Not used in NG9-1-1 CLDXF
- PN – Post (Pole) Number: Not used in NG9-1-1 CLDXF
- POBOX – Post Office Box: Not used in NG9-1-1 CLDXF
- RDSEC – Road Section: Not found in US addresses
- RDBR – Road Branch: Not found in US addresses
- RDSUBBR – Road sub-branch: Not found in US addresses

2.5.4 Follow a Tiered Approach to Inspecting LVF Datasets

When an LVF validates an address, it shall leverage a tiered approach to finding matches within its GIS data. It shall always attempt validation by inspecting its highest-accuracy GIS data first, starting with site/structure address points, if available and using subaddress first, followed by address points that do not have subaddress elements associated, and only fall back to less-accurate data, such as Road Centerlines when it cannot find a match in its higher-accuracy data. While an ECRF may choose to be more flexible in the data it attempts to use at the time of routing a call, using different datasets or changing the order in which they are searched, an LVF shall always examine its site/structure address points before its road centerlines. If the LVF cannot find one and only one site/structure address match when validating an address, it may fall back to searching for one and only one road centerline match.

The following examples illustrate how the number and type of GIS data matches specify the validity of a location:

SSAP matches	RCL matches	Valid?	Explanation
0	0	No	No GIS features matched the civic location.
0	1	Yes	One and only one road centerline matched the civic location.
0	2	No	The LVF cannot determine which feature is the true match.

SSAP matches	RCL matches	Valid?	Explanation
1	N/A	Yes	One and only one site/structure address point matched the civic location. The LVF considers the address is valid without attempting to discover matching road centerlines.
2	0	No	The LVF cannot determine which feature is the true match.
2	1	Yes	The LVF cannot determine within the site/structure address points layer which feature is a true match, but within the road centerline layer it can, therefore the response is valid.
2	2	No	The LVF cannot determine a single matching feature within either layer.

2.5.5 The LVF Shall Only Validate Against CLDXF-Compliant Data

To eliminate potential inconsistency between LVFs that utilize either legacy addressing data, or CLDXF-compliant data [8], or a combination of both, it is the intention of this document that an LVF only validate against CLDXF-compliant road centerline and site/structure address point data. For additional information, see section 2.6.2.

2.5.6 Support Special Characters

The LVF should expect to receive special characters within civic locations. Per RFC 5222 [7], civic locations in LoST requests are encoded using UTF-8, but not including pictographic characters.

2.5.7 Empty Elements Must Match Empty Record Values

If a civic location element in a PIDF-LO is empty in the LoST request and has no value, but with the XML tags (e.g., <tag><\tag>), the LVF must find a record that has a corresponding empty value. The presence of an empty civic location element in a proffered PIDF-LO indicates that the element is known to be "blank" and the LVF must find a match for the empty value for the element to be considered valid. If any PIDF-LO elements are omitted vs. empty, it should not be tested for validation unless it is considered a required element.

For example, consider a situation in which the LVF contains three site/structure address points, representing an industrial facility with two buildings and a main office. If a civic

location such as the one below is validated against this data, it will match the address point with the <null> Building attribute, since the civic location's <BLD> element has an empty value. However, if the <BLD> element had been omitted from the PIDF-LO altogether, the LVF would claim the civic location is invalid because it could not find a single matching address point. All three address points have different Building attribute values, so a BLD value in PIDF-LO is required to disambiguate amongst them.

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <country>US</country>
  <A1>TX</A1>
  <A2>WALLER COUNTY</A2>
  <RD>INDUSTRIAL</RD>
  <STS>AVENUE</STS>
  <HNO>4000</HNO>
  <PC>77355</PC>
  <BLD></BLD>
  <PCN>MAGNOLIA</PCN>
</civicAddress>
```

GIS Attribute Name	GIS Attribute Value
Country	US
State	TX
County	WALLER COUNTY
Inc_Muni	MAGNOLIA
StreetName	INDUSTRIAL
St_PosTyp	AVENUE
Add_Number	4000
Building	A
Post_Code	77355
Post_Comm	MAGNOLIA

GIS Attribute Name	GIS Attribute Value
Country	US
State	TX
County	WALLER COUNTY
Inc_Muni	MAGNOLIA
StreetName	INDUSTRIAL
St_PosTyp	AVENUE
Add_Number	4000
Building	B
Post_Code	77355
Post_Comm	MAGNOLIA

GIS Attribute Name	GIS Attribute Value
Country	US
State	TX
County	WALLER COUNTY
Inc_Muni	MAGNOLIA
StreetName	INDUSTRIAL
St_PosTyp	AVENUE
Add_Number	4000
Building	<null>
Post_Code	77355
Post_Comm	MAGNOLIA

Note: Attributes listed in PIDF-LO element order

2.5.8 House Number Validated Against Road Centerlines is Always "unchecked"

If a proffered civic location fails to match a site/structure address point, but does match a road centerline, the LVF will always consider the address' house number as unchecked. The LVF accordingly inserts the HNO element into the <unchecked> list within the resulting

findServiceResponse. The LVF considers a road centerline to match a civic location when its attribute values match the proffered location's civic location elements (subject to the exceptions listed previously), and its house number falls within its address ranges. Because the LVF can only claim that the address falls within the range of allowable addresses on the matching road centerline and cannot guarantee that the address actually exists in the real world, the LVF states that the house number is unchecked to indicate this fact. The only condition in which an LVF may claim the house number is valid is if the address matches a site/structure address point.

For example, consider a situation in which an LVF contains a road centerline whose address ranges include the house number of a proffered civic location that matches it. No site/structure address points exist that match the address.

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr">
  <country>US</country>
  <A1>MN</A1>
  <A2>HENNEPIN COUNTY</A2>
  <RD>KIRKWOOD</RD>
  <STS>CIRCLE</STS>
  <POD>NORTH</POD>
  <HNO>8979</HNO>
  <PC>55369</PC>
  <PCN>OSSEO</PCN>
</civicAddress>
```

All elements in the civic location are returned as valid except for the house number, which is returned as unchecked, because the LVF cannot determine if the address truly exists in the real world.

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:RD ca:STS ca:POD ca:PC ca:PCN</valid>
  <invalid></invalid>
  <unchecked>ca:HNO</unchecked>
</locationValidation>
```

GIS Attribute Name	GIS Attribute Value
FromAddr_L	8900
ToAddr_L	8998
FromAddr_R	8901
ToAddr_R	8999
Parity_L	E
Parity_R	O
StreetName	KIRKWOOD

GIS Attribute Name	GIS Attribute Value
St_PosTyp	CIRCLE
St_PosDir	NORTH
MSAGComm_L	MAPLE GROVE
MSAGComm_R	MAPLE GROVE
Country_L	US
Country_R	US
State_L	MN
State_R	MN
County_L	HENNEPIN COUNTY
County_R	HENNEPIN COUNTY
IncMuni_L	MAPLE GROVE
IncMuni_R	MAPLE GROVE
PostCode_L	55369
PostCode_R	55369
PostComm_L	OSSEO
PostComm_R	OSSEO

Note: Attributes not listed in PIDF-LO element order

Had the proffered civic location contained a house number outside of the road centerline's range:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <country>US</country>
  <A1>MN</A1>
  <A2>HENNEPIN COUNTY</A2>
  <RD>KIRKWOOD</RD>
  <STS>CIRCLE</STS>
  <POD>NORTH</POD>
  <HNO>9000</HNO>
  <PC>55369</PC>
  <PCN>OSSEO</PCN>
</civicAddress>
```

The following result would be returned, indicating the house number as invalid:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:RD ca:STS ca:POD</valid>
  <invalid>ca:HNO</invalid>
```

<unchecked>ca:PC ca:PCN</unchecked>
</locationValidation>

2.5.9 Do Not Use Alias Names When Validating an Address

Alias names shall not be used when determining if a civic location is valid. LVF records may have street name aliases, and it is important that their aliases not influence location validation to ensure consistent address representations in the LIS and mitigate potential issues such as address duplication. As such, valid civic locations shall match only the primary street name associated with an LVF record, and a proffered civic location that matches a street name alias instead of the record's primary street name shall not be considered valid. Note that an LVF could leverage the IETF similar locations draft (draft-ietf-ecrit-similar-location-03 [10]) to return, as a suggestion, a valid address containing the primary street name. The Alias name can be leveraged to make the correlation.

2.5.10 A Minimum Set of Elements Is Required To Attempt Validation

The i3 standard [2] leverages CLDXF [8] as a way to represent civic locations in PIDF-LO within the United States. CLDXF specifies a subset of civic location elements that must always be populated with values for a PIDF-LO civic locations to be CLDXF-compliant. As such, the LVF shall adhere to the CLDXF standard when validating civic locations. A proffered civic location must contain, at a minimum, values in the following civic locations elements for the LVF to attempt validation:

PIDF-LO Element Name	CLDXF Element Name
country	Country
A1	State
A2	County
A3	Incorporated Municipality
RD or LMK/LMKP	Either Street Name or Complete Landmark Name/Landmark Name Part

CLDXF defines other elements as conditional, and they may be omitted from PIDF-LO in cases where the information is not needed to represent a valid civic location. For example, if a landmark has no street address, the CLDXF standard requires that the PIDF-LO contains LMKP(s) and LMK but may omit RD and HNO. Likewise, for a street address that is not a landmark, RD and HNO are required and LMKP(s) and LMK may be omitted. For further information on conditional elements and the conditions that govern their inclusion or omission, see section 3 of the CLDXF standard.

2.6 Operational Recommendations

2.6.1 Recommendations for GIS Data Providers

Local GIS Authorities and GIS Provisioning Authorities play a key role in maintaining GIS data for use in location validation and making it available for use. These GIS providers should consider the following recommendations when maintaining data and provisioning it to an LVF:

- If an LVF did not have site/structure address point data for a given geographic region, locations in that region may have not been validated with a high degree of accuracy. For example, locations may have had subaddress elements that were not validated because road centerlines do not include subaddress attributes. A LIS operator may have chosen to accept the locations and insert them into the LIS, even though some of the locations' address elements were unchecked. If, at some future time, site/structure address point data within the geographic region was provisioned to the LVF, the locations that were previously inserted into the LIS may be rendered invalid in the presence of the more-detailed address data. As such, whenever GIS data is provisioned to an LVF that is higher accuracy and/or contains more detail than the GIS data that was provisioned previously, any records in the LIS that fall within the updated geography should be revalidated, ideally leveraging the IETF "planned changes" document [11]. The GIS Data Provider should notify any LVF Clients of the update, so that they can schedule a revalidation of records within their databases.
- The GIS Data Provider should be aware that this document recommends in section 2.4.1.2 that feature attribution alone is utilized during validation and geometry for validation is not used, though this may be revisited in a future revision of this document.

2.6.2 Recommendations for LVF Operators

LVF Operators should consider the following recommendations when implementing, supporting, and maintaining an LVF:

- Ensure that the LVF is configured to use CLDXF-compliant fields within its GIS data. Some GIS data layers may contain a mixture of attribute fields, some of which represent address components that are MSAG-valid and some that are CLDXF-compliant. The LVF should not be configured to use legacy attribute fields. For example, if the LVF allows its GIS data fields to be mapped to known PIDF-LO address elements, the CLXDF-compliant Incorporated Municipality field should be mapped to A3, not MSAG Community.
- Ensure that the LVF only utilizes authoritative GIS data such as road centerlines and site/structure address points when validating addresses, per section 2.5.1. While

some vendor-specific implementations of the LVF may allow for the use of alternative types of address data, using it for validation purposes is strongly discouraged due to the potential inaccuracy of the data and the greatly-increased risk of inconsistent behavior from other LVFs. For example, an LVF Operator may have originally procured an LVF that allowed the use of a MSAG to validate addresses. If the LVF Operator ceases the use of their current LVF and begins using an LVF from a different vendor, the new LVF may follow the guidelines in section 2.5.1 and only use road centerline and site/structure address points to validate addresses. The new LVF will likely produce different validation results than the first LVF, resulting in a potentially large amount of data remediation work to fix issues reported by the new LVF.

- For additional recommendations, please see section 2.3.5 referencing the use of LoST extensions.

2.6.3 Recommendations for LVF Clients

The recommendations in this section are intended to provide direction that should help LVF Clients (e.g. Origination Network and Service Providers, and Terminating ESRPs) with the goal of maximizing successful revalidations should LVF vendors change within a given region.

2.6.3.1 Leveraging IETF “Complete and Similar Location Info” Document

It is expected that the “similar location” document [10] will reach RFC status and will be required to implement at the LVF. It is recommended that both the LVF and the LVF clients implement it now.

2.6.3.1.1 Leveraging Completed Location Returned with a VALID response

In the case of a valid response, when using the similar location extension of LoST, the complete record that the request validated against, with all associated fields, will be returned in the LoST response. That complete record should be placed in the LIS, rather than only provisioning the address elements used in the LoST request. This is because a given address may be deemed valid, even if it is incomplete as compared to the record it validated against.

For example:

GIS Attribute Name	Input PIDF-LO Value	Output PIDF-LO Value
Country	US	US
State	TX	TX
County	WALLER COUNTY	WALLER COUNTY

Inc_Muni	MAGNOLIA	MAGNOLIA
StreetName	INDUSTRIAL	INDUSTRIAL
St_PosTyp		AVENUE
Add_Number	4000	4000
Post_Code	77355	77355
Post_Comm		MAGNOLIA

Note, the order presented here is the sequence order of the elements in the PIDF-LO, not to be confused with the hierarchical order of validation described in section 2.5.3. In this example, St_PosTyp and Post_Comm were not included in the LoST validation request to the LVF. Using the Similar Location extension to LoST, the site/structure address point record that the location validated against is returned in the response. When the LIS operator places this subscriber record in their LIS, both St_PosTyp and Post_Comm should also be included in the LIS record.

2.6.3.1.2 Leveraging Similar Location(s) Returned with an INVALID response

From the client side, it is expected that it will take a human to ultimately determine whether the similar locations provided in a response represent the actual address.

Future work will be needed to help ensure that similar locations returned in an LVF response will occur in a consistent manner.

2.6.3.2 Examine the Service Mappings

The LVF can be used for more than location validation; it can also be used to uncover data problems. LVF clients should not only examine LVF responses to determine the validity of a civic location – they should, when possible, examine the service mapping(s) contained within the response to determine if the correct service was returned. Such examination can provide a way to uncover potential call routing problems, for example, when validating civic locations using the urn:nena:service:sos.psap service URN. Should a problem be discovered, it should be resolved through cooperation with the LVF Operator and/or 9-1-1 Authority.

2.6.3.3 Perform a Validation Gap Analysis When Switching LVF Vendors

If an LVF Client plans to use an LVF supplied or operated by a different LVF Operator than their current LVF, the client should perform a batch validation of their location database records against both LVFs. Doing so will allow the LVF Client to determine if any data remediation work exists before switching to a new LVF Operator, or if any performance or reliability differences exist with the different LVF Operator that could impact regular or batch validation operations. Should gaps be discovered, the LVF Client should cooperate with the GIS Data Authority and/or 9-1-1 Authority to plan any remediation activities.

The performance of an LVF is not specified in this document. There is no defined batch verification method. A large number of validation requests from a single source in a small time period looks like a Denial of Service (DoS) attack and may be treated as such.

If the LVF vendors changes, the LVF client should revalidate all the LIS entries, even though the goal of this document is to assure that both the old and new vendors give the same results. Such a revalidation would need to be carefully managed so that the result doesn't trip DoS protections. It should be noted that without extraordinary cooperation between vendors (possibly required by contract), the new LVF would get the URLs stored by the old LVF for the lost-planned-changes notification mechanism, which by itself would trigger a mass revalidation on the new LVF.

2.6.3.4 Leverage LoST Planned Changes

It is expected that the "planned changes" document [11] will reach RFC status and will be required to implement at the LVF. It is recommended that both the LVF and LVF clients implement it now. Note that a change that affects a large number of locations will cause a large number of revalidations at the LVF. This may look like a DoS attack. The LVF (or the provisioning source) could ameliorate this problem by, for example, staggering the announcement of the change so that the revalidations were spread over time. Consideration of this issue may influence both the implementation and the requirements expressed in an LVF procurement.

3 Impacts, Considerations, Abbreviations, Terms, and Definitions

3.1 Operations Impacts Summary

Beyond the initial effort required to create and maintain GIS data provisioned for use by an LVF and use of the LVF for civic address validation prior to LIS subscriber record provisioning, the recommendations in this document will reduce the operational impacts on all stakeholders. Failure to follow the recommendations in this document would in turn increase the amount of time required by all parties to resolve any inconsistencies and make subsequent modifications to the underlying data. This increase in time to resolve inconsistencies could result in increased costs or delays in deployment of NG9-1-1 systems.

Examples of impacts that might be experienced by not following the recommendations in this document include:

- Increase in the cost and time requirements placed upon GIS Data Providers and Originating Service Providers to resolve civic locations that were previously valid in LVF A but are invalid in LVF B. This includes data management processes and procedures that could be negatively impacted.
- Delay the switch from LVF A to LVF B or delay the deployment of multiple LVFs provided by different LVF operators.

3.2 Technical Impacts Summary

NG9-1-1 relies on location. The processes described within this document help ensure location consistency between systems that utilize location. The recommendations in this document better enable systems to treat location information programmatically as opposed to requiring human intervention to resolve location validation problems. Overall, following the recommendations in this document will result in improved and consistent location-based routing.

3.3 Security Impacts Summary

This document does not introduce any security impacts above those introduced in NENA-STA-010 [2].

3.4 Recommendation for Additional Development Work

Recommendations for future work within this document:

Section	Reference to future work
2.4.1.3	To mitigate the potential inconsistency described in section 2.4.1.3, future work should recommend or standardize the approaches used to determine the service mapping(s) to return in validation responses. Some LVF implementations employ geocoding procedures that interpolate points along road centerlines, while other LVF implementations may perform, for example, a spatial intersection query against a service boundary layer using the entire road centerline geometry. The variance in these approaches can cause LVFs from different vendors or operators to produce inconsistent results.
<general>	Determine if the LVF should convey a level of accuracy to the Public Safety Answering Point (PSAP). Consider if the LVF should return a degree of "accuracy" in a location validation response, and if the LIS should store that, so that it could be conveyed to a PSAP at call time.
2.4.1.2 & 2.6.1	Determine if the LVF should perform spatial analysis of the relationship between a civic location and administrative boundaries (state, county, Incorporated Municipality, Unincorporated Community, Neighborhood Community), when validating locations.
2.4.1.5	There is a strong need to address transitional considerations for LVF vendor implementations. One of many examples, is the need to address the Master Street Address Guide (MSAG)

	Community element use during transition to an i3 end state.
2.6.3.1	Once draft-ietf-ecrit-similar-location [10] has been accepted by the Internet Engineering Task Force (IETF) as a Request for Comments (RFC), continue to elaborate on how similar locations can be consistently returned between LVF Operators.

Recommendations for future work in other documents:

Document	Reference to future work
NENA-STA-005 [3]	Describe and mitigate any variance in how GIS data is supplied to the SI by GIS Data Providers and what a SI Operator does to the data to prepare it for the LVF that could cause unexpected and inconsistent behaviors in different LVF systems.
NENA STA-006 [9]	Recommending the inclusion of “do not use for validation” flags for road centerlines, one per side, where the 9-1-1 authority wishes to use site/structure address points only, excluding certain road centerline records.
NENA-STA-010 [2] or IETF	Develop a LoST extension to return which GIS dataset/layer was used to validate a location.
draft-ietf-ecrit-similar-location [10]	Continued development of an extension to LoST (RFC 5222 [7]) to return complete and similar location information.
draft-itef-ecrit-lost-planned-changes [11]	Continued development of an extension to LoST (RFC 5222 [7]) to allow for validation of locations around a planned change.
<general>	Identify and document technical and operational considerations regarding the interconnection of LVF Clients (such as those used by Service Providers) with LVFs provided by LVF Operators.

3.5 Anticipated Timeline

The timeline for implementation and deployment of the recommendations outlined within this document may vary based upon stakeholders' ability to employ or adhere to this document's recommendations.

3.6 Cost Factors

Following this document will achieve standardization between LVFs results. This consistency will have expected long-term cost savings and reduce transitional costs. Some stakeholders that will enjoy cost savings include LVF Operators, LVF Clients, and GIS Data Authorities. However, achieving this consistency itself does not come for free. Several stakeholders will bear the cost of following this document's recommendations to ensure that LVF systems validate locations consistently. These costs will impact LVF Providers, LVF Operators, and GIS Data Authorities. LVF Providers may need to augment the design of their systems to bring them in line with this document's recommendations. LVF Operators that are currently utilizing LVFs may need to expend time and resources to assist in upgrading their software to newer versions that adhere to this document or may need to reconfigure their systems to ensure they behave in the ways outlined within this document. As per the recommendations in this document, GIS Data Authorities will also need to devote time and resources to ensuring their GIS data is compliant with the Civic Location Data eXchange Format (CLDXF) standard [8] and the NG9-1-1 GIS Data Model [9]. These cost factors are in addition to the more general costs that stakeholders may bear, such as ongoing systems engineering and standards development for new features and functionality, ongoing customer management, and systems administration and operational costs.

If the work to follow the recommendations outlined in this document is not performed, and LVF systems do not validate locations consistently, LVF Operators, LVF Clients, and GIS Data Authorities may bear much more cost when working with LVFs from different providers or replacing an LVF with one from a different provider, as described in section 2.1. To minimize long-term cost, it is recommended that affected stakeholders adopt the recommendations put forth in this document before bearing the cost of LVF inconsistency.

3.7 Cost Recovery Considerations

GIS Data Authorities may be able to recover some, or all, of the cost of ensuring that GIS datasets adhere to the recommendations outlined within this document through state or regional GIS grant programs. However, additional cost recovery mechanisms do not appear to exist for other stakeholders and the work they must perform to ensure LVF consistency. Normal business practices shall be assumed to be the cost recovery mechanism in many, or all, of these cases. If the work to follow the recommendations outlined in this document is not performed, and LVF systems do not validate locations consistently, multiple parties will face additional costs.

3.8 Additional Impacts (non-cost related)

The information contained in this NENA document is expected to have both 9-1-1 technical and 9-1-1 Center operations impacts, based on the analysis of the authoring group. At the date of publication of this document, development had not started. The primary impacts are expected to include:

Creation of uniform validation standards within the 9-1-1 community that provides incentive for entities not directly connected to the 9-1-1 system to conform to those standards and avoids balkanization of representing addresses which may interact at some point with 9-1-1.

3.9 Abbreviations, Terms, and Definitions

See NENA Master Glossary of 9-1-1 Terminology, NENA-ADM-000 [1], for a complete listing of terms used in NENA documents. All abbreviations used in this document are listed below, along with any new or updated terms and definitions.

Term or Abbreviation (Expansion)	Definition / Description
<i>9-1-1 Authority</i> <i>AKAs: Authority Having Jurisdiction (AHJ), 9-1-1 Governing Authority, 9-1-1 Administrator</i>	A State, County, Regional or other governmental entity responsible for 9-1-1 service operations. For example, this could be a county/parish or city government, a special 9-1-1 or Emergency Communications District, a Council of Governments or other similar body.
<i>ALI (Automatic Location Identification)</i>	A function in the E9-1-1 system that provides customer name, the caller's telephone number, and address/location information plus other related supplementary information for display at the PSAP for a given 9-1-1 call.
<i>Alternate Address Record</i>	An Alternate Address record may be the Postal equivalent to the MSAG address or it may be any other alternate address provided (i.e. an alias street name – John Carpenter Freeway vs. Highway 121).
<i>Border Control Function (BCF)</i>	Provides a secure entry into the ESInet for emergency calls presented to the network. The BCF incorporates firewall, admission control, and may include anchoring of session and media as well as other security mechanisms to prevent deliberate or malicious attacks on PSAPs or other entities connected to the ESInet.

<i>CLDXF (Civic Location Data Exchange Format)</i>	A United States emergency services profile of PIDF-LO that defines a set of standard data elements that describe detailed street address information. See NENA-STA-004 [8].
<i>Customer Premises Equipment (CPE)</i>	Communications or terminal equipment located in the customer's facilities – Terminal equipment at a PSAP.
<i>DoS (Denial of Service)</i>	A type of cyber-attack intended to overwhelm the resources of the target and deny the ability of legitimate users of the target the normal service the target provides. NENA-INF-015 NG9-1-1 Security Information Document DDoS (Distributed Denial of Service Attack) A distributed denial-of-service (DDoS) is where the attack source is more than one, often thousands of, unique IP addresses. A DDoS attack occurs when multiple systems flood the bandwidth or resources of a targeted system, usually one or more web servers. Such an attack is often the result of multiple compromised systems (for example a botnet) flooding the targeted system with traffic. NENA-INF-015 NG9-1-1 Security Information Document TDoS (Telephone Denial of Service) Illegal attacks targeting the telephone network by generating numerous 9-1-1 phone calls, tying up the network and preventing an agency from receiving legitimate calls. NENA-INF-015 NG9-1-1 Security Information Document
<i>DHCP (Dynamic Host Control Protocol (i2) Dynamic Host Configuration Protocol)</i>	A widely used configuration protocol that allows a host to acquire configuration information from a visited network and, in particular, an IP address.
<i>DNS (Domain Name Server (or Service or System))</i>	Used in the Internet today to resolve domain names. The input to a DNS is a domain name (e.g., 67elcordia.com); the response is the IP address of the domain. The DNS allows people to use easy to remember text-based addresses and the DNS translates those names into routable IP addresses.

<i>E9-1-1 (Enhanced 9-1-1)</i>	A telephone system which includes network switching, database and Public Safety Answering Point premise elements capable of providing automatic location identification data, selective routing, selective transfer, fixed transfer, and a call back number. The term also includes any enhanced 9-1-1 service so designated by the Federal Communications Commission in its Report and Order in WC Docket Nos. 04-36 and 05-196, or any successor proceeding.
<i>Emergency Call Routing Function (ECRF)</i>	A functional element in an ESInet which is a LoST protocol server where location information (either civic address or geo-coordinates) and a Service URN serve as input to a mapping function that returns a URI used to route an emergency call toward the appropriate PSAP for the caller's location or towards a responder agency.
<i>ECRF/LVF</i>	While the ECRF and LVF are separate functional elements, the GIS data set provisioned to each is expected to be the same. Where used, the ECRF/LVF designation indicates the terms are interchangeable.
<i>ecrit (Emergency Context Resolution with Internet Technologies)</i>	An Internet Engineering Task Force (IETF) Working Group whose focus is emergency services protocols and mechanisms.
ESN (Emergency Service Number)	A 3-5 digit number that represents one or more ESZs. An ESN is defined as one of two types: Administrative ESN and Routing ESN.
<i>ESRD (Emergency Services Routing Digits)</i>	A 10-digit North American Numbering Plan number that uniquely identifies a base station, cell site, or sector that may be used to route wireless emergency calls through the network. The ESRD may also be used by the PSAP to retrieve the associated ALI data.
<i>ESRK (Emergency Services Routing Key)</i>	A 10-digit North American Numbering Plan number that uniquely identifies a wireless emergency call and may be used to route the call through the emergency services network, and is used to retrieve the associated ALI data.
<i>FGDC (Federal Geographic Data Committee)</i>	The Federal Geographic Data Committee (FGDC) is an interagency committee that promotes the coordinated development, use, sharing, and dissemination of geospatial data on a national basis. https://www.fgdc.gov/

<i>GIS (Geographic Information System)</i>	A system for capturing, storing, displaying, analyzing and managing data and associated attributes which are spatially referenced.
<i>GIS Data Provider</i>	Provides authoritative GIS data to systems that utilize it, such as ECRF/LVF, via the Spatial Interface (SI).
<i>NG9-1-1 Functional Element Operator</i>	An entity that operates one or more of the functional elements as described in NENA standards, including STA-010.
<i>IETF (Internet Engineering Task Force)</i>	Global standards development organization for the development of Internet protocols.
<i>Incumbent Local Exchange Carrier (ILEC)</i>	A telephone company that had the initial telephone company franchise in an area.
<i>LIS (Location Information Server)</i>	A Location Information Server (LIS) is a functional element in an IP-capable originating network that provides locations of endpoints (i.e., calling device). A LIS can provide location by reference or by value, in geodetic or civic forms.
<i>LIS Operator</i>	An entity that manages and maintains the LIS associated with the IP access network that is used in emergency calls.
<i>LoST (Location-to-Service Translation) Protocol</i>	A protocol that associates locations and service (Uniform Resource Names (URNs) with service Uniform Resource Identifiers (URIs). Used for location-based call routing and location validation. In NG9-1-1, used as the protocol for the ECRF and LVF.
<i>LVF (Location Validation Function)</i>	A Next Generation 9-1-1 Core Services (NGCS) functional element that uses the LoST protocol to validate civic location information against the authoritative GIS database information.
<i>LVF Operator</i>	An entity that manages and maintains LVF LoST Servers.
<i>Master Street Address Guide (MSAG)</i>	A database of street names and house number ranges within their associated communities defining Emergency Service Zones (ESZs) and their associated Emergency Service Numbers (ESNs) to enable proper routing of 9-1-1 calls.

NG9-1-1 (Next Generation 9-1-1)	NG9-1-1 is an Internet Protocol (IP)-based system comprised of managed Emergency Services IP networks (ESInets), functional elements (applications), and databases that replicate traditional E9-1-1 features and functions and provides additional capabilities. NG9-1-1 is designed to provide access to emergency services from all connected communications sources, and provide multimedia data capabilities for Public Safety Answering Points (PSAPs) and other emergency service organizations. www.nena.org/resource/resmgr/ng9-1-1_project/whatisng911.pdf NOTE: It is recognized that there will be a multi-year transition to NG9-1-1 beginning as early as 2010. See the NENA list of FAQs related to NG9-1-1 for more details.
<i>NGCS (Next Generation 9-1-1 Core Services)</i>	The base set of services needed to process a 9-1-1 call on an ESInet. Includes the Emergency Services Routing Proxy (ESRP), ECRF, LVF, Border Control Function (BCF), Bridge, Policy Store, Logging Services and typical IP services such as Domain Name System (DNS) and Dynamic Host Configuration Protocol (DHCP). The term NG9-1-1 Core Services includes the services and not the network on which they operate.
NENA (National Emergency Number Association)	The National Emergency Number Association is a not-for-profit corporation established in 1982 to further the goal of "One Nation-One Number." NENA is a networking source and promotes research, planning and training. NENA strives to educate, set standards and provide certification programs, legislative representation and technical assistance for implementing and managing 9-1-1 systems.
<i>Origination Network</i>	The network which originates a 9-1-1 call. Includes the access network and the calling network. Typically operated by carriers or other service providers.
<i>pANI (Pseudo Automatic Number Identification)</i> <i>AKA: routing number</i>	A telephone number used to support routing of wireless 9-1-1 calls. It may identify a wireless cell, cell sector or PSAP to which the call should be routed.
<i>PIDF-LO (Presence Information Data Format – Location Object)</i>	A standard structure that represents location information defined by an Extensible Markup Language (XML) schema.
<i>PSAP (Public Safety Answering Point)</i>	An entity responsible for receiving 9-1-1 calls and processing those calls according to specific operational policies.

<i>RFC (Request for Comment)</i>	An Internet protocol standard published by the IETF.
<i>Service Provider</i>	An entity providing one or more of the following 9-1-1 elements: network, Customer Premise Equipment (CPE), or database service
<i>SI (Spatial Interface)</i>	A standard interface used to provision GIS data to ECRF/LVF or other systems.
<i>SI Operator</i>	An entity that manages and maintains the SI.
<i>SOI (Service Order Input)</i>	A file of completed service order updates that is sent to the Database Management System Provider by all Service Providers.
<i>Telephone Number (TN)</i>	A sequence of digits assigned to a device to facilitate communications via the public switched telephone network or other private network.
<i>TSP (Telecommunications Service Provider)</i>	A business that provides voice or data transmission services. These services are provided over a telecommunications network that transmits any combination of voice, video and/or data between users. A TSP could be, but is not limited to, a Local Exchange Carrier (LEC), a wireless telecommunications provider, a Commercial Mobile Radio Service provider, or a PBX service provider.
<i>Uniform Resource Identifier (URI)</i>	A predictable formatting of text used to identify a resource on a network (usually the Internet) OR A string of characters that must follow prescribed syntaxes such as URL, URN... Note Version 1.1 of the XML namespaces recommendation uses IRIs (Internationalized Resource Identifiers) instead of URIs. However, because version 1.1 is not yet a full recommendation [February, 2003] and because the IRI RFC is not yet complete, this document continues to refer to URIs instead of IRIs.
<i>Uniform Resource Name (URN)</i>	Uniform Resource Identifiers (URIs) that use the URN scheme, and are intended to serve as persistent, location-independent resource names.
<i>VPC (VoIP Positioning Center)</i>	The element that provides routing information to support the routing of VoIP emergency calls, and cooperates in delivering location information to the PSAP over the existing ALI DB infrastructure.
<i>VSP (VoIP Service Provider)</i>	A company that offers VoIP telecommunications services that may be used to generate a 9-1-1 call, and interconnects with the emergency services network.

<i>extensible Markup Language (XML)</i>	An internet specification for web documents that enables tags to be used that provide functionality beyond that in Hyper Text Markup Language (HTML). Its reference is its ability to allow information of indeterminate length to be transmitted to a PSAP call taker or dispatcher versus the current restriction that requires information to fit the parameters of pre-defined fields.
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4 Recommended Reading and References

- [1] NENA Master Glossary of 9 1 1 Terminology, [NENA-ADM-000](#)
- [2] National Emergency Number Association. "NENA Detailed Functional and Interface Standards for the NENA i3 Solution" NENA-STA-010. Posted at: http://www.nena.org/?page=i3_Stage3
- [3] National Emergency Number Association. "NENA Standards for the Provisioning and Maintenance of GIS data to ECRFs and LVFs" NENA-STA-005. Posted at: <http://www.nena.org/page/ProvGISECRFLVF>
- [4] Internet Engineering Task Force. "A Presence-based GEOPRIV Location Object Format" RFC 4119 Posted at: <https://tools.ietf.org/html/rfc4119>
- [5] Internet Engineering Task Force. "Revised Civic Location Format for Presence Information Data Format Location Object (PIDF-LO)" RFC 5139 Posted at: <https://tools.ietf.org/html/rfc5139>
- [6] Internet Engineering Task Force. "GEOPRIV Presence Information Data Format Location Object (PIDF-LO) Usage Clarification, Considerations, and Recommendations" RFC 5491 Posted at: <https://tools.ietf.org/html/rfc5491>
- [7] Internet Engineering Task Force. "LoST: A Location-to-Service Translation Protocol" RFC 5222 Posted at: <https://tools.ietf.org/html/rfc5222>
- [8] National Emergency Number Association. "NENA Next Generation 9-1-1 United States Civic Location Data Exchange Format (CLDXF)" NENA-STA-004. Posted at: <http://www.nena.org/?NG911CLDXF>
- [9] National Emergency Number Association. "NENA Standard for NG9-1-1 GIS Data Model" NENA-STA-006 (Work in Progress DRAFT) Posted at: https://dev.nena.org/kws/public/download/9828/20161206_NG9-1-1%20GIS%20Data%20Model_PubRvw.pdf
- [10] Internet Engineering Task Force. "A LoST extension to return complete and similar location info" draft-ietf-ecrit-similar-location (Work in Progress DRAFT) Posted at: <https://tools.ietf.org/html/draft-ietf-ecrit-similar-location>
- [11] Internet Engineering Task Force. "Validation of Locations Around a Planned Change" draft-ecrit-lost-planned-changes (Work in Progress DRAFT) Posted at: <https://tools.ietf.org/html/draft-ecrit-lost-planned-changes>

- [12] National Emergency Number Association. "NENA Next Generation 9-1-1 Data Management Requirements" NENA-REQ-002. Posted at:
<https://www.nena.org/page/NGDataMgmt>

Appendix A – Examples of LVF Inconsistency

This appendix contains a set of illustrative examples which highlight cases of potential behavioral inconsistency between LVF implementations. The examples reference recommendations to improve or ensure consistency that appear earlier in this document.

A.1 Lack of Minimum Element Requirements

Section 2.4.1.1 details a problem that can lead to LVF inconsistency, when LVFs do not follow recommendations that pertain to minimum element requirements for civic locations expressed in

PIDF-LO. This example illustrates the possible differing behavior of two LVFs. Given this civic location and a site/structure address point in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <country>US</country>
  <A2>WALLER COUNTY</A2>
  <RD>WOODWAY</RD>
  <STS>DRIVE</STS>
  <HNO>137</HNO>
  <PC>77355</PC>
  <PCN>MAGNOLIA</PCN>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country</valid>
  <invalid>ca:A1</invalid>
  <unchecked>ca:A2 ca:RD ca:STS ca:HNO ca:PCN ca:PC </unchecked>
</locationValidation>
```

LVF B might return the following, different result:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A2 ca:RD ca:STS ca:HNO ca:PCN ca:PC </valid>
  <invalid></invalid>
  <unchecked></unchecked>
</locationValidation>
```

A summary of the proffered civic address, its matching site/structure address point, and the results from LVF A and LVF B are as follows:

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country	US	<country>US</country>	Valid	Valid
A1	State	TX		Invalid	
A2	County	WALLER	<A2>WALLER	Unchecked	Valid

		COUNTY	COUNTY</A2>		
A3	Inc_Muni	MAGNOLIA		Unchecked	
RD	StreetName	WOODWAY	<RD>WOODWAY</RD>	Unchecked	Valid
STS	St_PosTyp	DRIVE	<STS>DRIVE</STS>	Unchecked	Valid
HNO	Add_Number	137	<HNO>137</HNO>	Unchecked	Valid
PC	Post_Code	77355	<PC>77355</PC>	Unchecked	Valid
PCN	Post_Comm	MAGNOLIA	<PCN>MAGNOLIA</PCN>	Unchecked	Valid

Note that elements in this table are provided in PIDF-LO element order.

From a baseline perspective, LVF A claims the civic location is invalid, but LVF B claims the location is valid. LVF A claims the location is invalid because it lacks an A1 value per section 2.3. However, LVF B claims the location is valid because it does not follow the guidelines in section 2.3, and is able to find a single record that it believes matches the location.

A.2 Use of Administrative Boundaries to Determine Validity

Section 2.4.1.2 details a problem that can lead to LVF inconsistency, if LVFs examine spatial relationships to administrative boundaries when validating locations. This example illustrates the possible differing behavior of two LVFs. Given this civic location and a site/structure address point in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <country>US</country>
  <A1>TX</A1>
  <A2>WALLER COUNTY</A2>
  <A3>MAGNOLIA</A3>
  <RD>WOODWAY</RD>
  <STS>DRIVE</STS>
  <HNO>137</HNO>
  <PC>77355</PC>
  <PCN>MAGNOLIA</PCN>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:A3 ca:RD ca:STS ca:HNO ca:PCN ca:PC</valid>
  <invalid></invalid>
  <unchecked></unchecked>
</locationValidation>
```

LVF B might return the following, different result:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1</valid>
```

<invalid>ca:A2</invalid>
<unchecked>ca:A3 ca:RD ca:STS ca:HNO ca:PCN ca:PC</unchecked>
</locationValidation>

A summary of the proffered civic address, its matching site/structure address point, and the results from LVF A and LVF B are as follows:

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country	US	<country>US</country>	Valid	Valid
A1	State	TX	<A1>TX</A1>	Valid	Valid
A2	County	WALLER COUNTY	<A2>WALLER COUNTY</A2>	Valid	Invalid
A3	Inc_Muni	MAGNOLIA	<A3>MAGNOLIA</A3>	Valid	
RD	StreetName	WOODWAY	<RD>WOODWAY</RD>	Valid	Unchecked
STS	St_PosTyp	DRIVE	<STS>DRIVE</STS>	Valid	Unchecked
HNO	Add_Number	137	<HNO>137</HNO>	Valid	Unchecked
PC	Post_Code	77355	<PC>77355</PC>	Valid	Unchecked
PCN	Post_Comm	MAGNOLIA	<PCN>MAGNOLIA</PCN>	Valid	Unchecked

Note that elements in this table are provided in PIDF-LO element order.

In this example, the address point happens to exist within the A2 (county) boundary for GRIMES COUNTY, not WALLER COUNTY. LVF A did not examine the spatial relationship of the address point to the administrative boundaries that it intersects, and only examined the attributes of the address points to search for a match. Since it found a match, LVF A claims the address is valid. LVF B, however, determined that the A2 (county) boundary that intersects the address point is not WALLER COUNTY – and thus claims the A2 element of the civic location is invalid.

A.3 Variance in Geocoding Offsets

Section 2.4.1.3 details a problem that can lead to LVF inconsistency, if LVFs support both geocoding civic locations along road centerlines, and the LVFs are configured with different side offsets. This example illustrates the possible differing behavior of two LVFs. Given this civic location and a road centerline in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr">
  <country>US</country>
  <A1>MT</A1>
  <A2>TOOLE COUNTY</A2>
  <A3>UNINCORPORATED</A3>
  <A4>SWEET GRASS</A4>
  <RD>CENTRAL</RD>
  <STS>AVENUE</STS>
  <HNO>270</HNO>
  <PC>59484</PC>
```

```
<PCN>SWEET GRASS</PCN>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:A3 ca:A4 ca:RD ca:STS ca:HNO ca:PCN ca:PC </valid>
  <invalid></invalid>
  <unchecked></unchecked>
</locationValidation>
```

LVF B might return the following, different result:

```
<?xml version="1.0" encoding="UTF-8"?>
<errors xmlns="urn:ietf:params:xml:ns:lost1" source="lost.nena.org">
  <notFound message="The server could not find an answer to the query." xml:lang="en"/>
</errors>
```

A summary of the proffered civic address, its matching road centerline, and the results from LVF A and LVF B are as follows, listed using PIDF-LO element ordering:

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country_L Country_R	US US	<country>US</country>	Valid	LoST Error
A1	State_L State_R	MT MT	<A1>MT</A1>	Valid	
A2	County_L County_R	TOOLE COUNTY TOOLE COUNTY	<A2>TOOLE COUNTY</A2>	Valid	
A3	IncMuni_L IncMuni_R	UNINCORPORATED UNINCORPORATED	<A3>UNINCORPORATED</A3>	Valid	
A4	UnincCom_L UnincCom_R	SWEET GRASS SWEET GRASS	<A4>SWEET GRASS</A4>	Valid	
RD	StreetName	CENTRAL	<RD>CENTRAL</RD>	Valid	
STS	St_PosTyp	AVENUE	<STS>AVENUE</STS>	Valid	
HNO	FromAddr_L ToAddr_L FromAddr_R ToAddr_R Parity_L Parity_R	236 298 237 299 E O	<HNO>270</HNO>	Valid	

	MSAGComm_L	SWEET GRASS			
	MSAGComm_R	SWEET GRASS			
PC	PostCode_L	59484	<PC>59484</PC>	Valid	
	PostCode_R	59484			
PCN	PostComm_L	SWEET GRASS	<PCN>SWEET GRASS</PCN>	Valid	
	PostComm_R	SWEET GRASS			

From a baseline perspective, LVF A claims the civic location is valid, but LVF B returns a LoST notFound error and is unable to return a validation result. In this example, LVF A has small geocoding offsets and the offsets used in LVF B are much greater. LVF A interpolates a geodetic point that falls within a service boundary, so it can validate the address and return a result. LVF B, however, interpolates a point that is far away from the road centerline – so far away that the interpolated point falls outside a service boundary, in an area where none exist – across the Canadian border. The LVF was not provisioned with the coverage region of a Canadian LVF and none existed that could validate the location through redirection by a Forest Guide. A LoST findServiceResponse must contain one or more service mappings, and because LVF B cannot find any to return, it instead returns a notFound error with no validation result at all.

A.4 Use of Alias Names to Determine Validity

Section 2.4.1.4 details a problem that can lead to LVF inconsistency, if LVFs attempt to use street alias names when validating civic locations. This example illustrates the possible differing behavior of two LVFs. Given this civic location and a site/structure address point in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr"
xmlns:cae="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr:ext">
  <country>US</country>
  <A1>MN</A1>
  <A2>HENNEPIN COUNTY</A2>
  <A3>MAPLE GROVE</A3>
  <RD>30</RD>
  <HNO>11852</HNO>
  <PC>55369</PC>
  <PCN>OSSEO</PCN>
  <cae:STP>COUNTY ROAD</cae:STP>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr"
xmlns:cae="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr:ext">
  <valid>ca:country ca:A1 ca:A2 ca:A3</valid>
  <invalid>ca:RD</invalid>
  <unchecked>ca:HNO ca:PCN ca:PC cae:STP</unchecked>
```

</locationValidation>

LVF B might return the following, different result:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr"
xmlns:cae="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr:ext">
  <valid>ca:country ca:A1 ca:A2 ca:A3 ca:RD ca:STS ca:HNO ca:PCN ca:PC cae:STP</valid>
  <invalid></invalid>
  <unchecked></unchecked>
</locationValidation>
```

A summary of the proffered civic address, its matching road centerline, and the results from LVF A and LVF B are as follows, listed using PIDF-LO element ordering:

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country	US	<country>US</country>	Valid	Valid
A1	State	MN	<A1>MN</A1>	Valid	Valid
A2	County	HENNEPIN COUNTY	<A2>HENNEPIN COUNTY</A2>	Valid	Valid
A3	Inc_Muni	MAPLE GROVE	<A3>MAPLE GROVE</A3>	Valid	Valid
STP	St_PreTyp		<cae:STP>COUNTY ROAD</cae:STP>	Unchecked	Valid
RD	StreetName	93RD	<RD>30</RD>	Invalid	Valid
STS	St_PosTyp	AVENUE			Valid
POD	St_PosDir	NORTH			
HNO	Add_Number	11852	<HNO>11852</HNO>	Unchecked	Valid
PC	Post_Code	55369	<PC>55369</PC>	Unchecked	Valid
PCN	Post_Comm	OSSEO	<PCN>OSSEO</PCN>	Unchecked	Valid

In this example, the address point has a street name alias of COUNTY ROAD 30. LVF A does not validate against street name aliases, and requires that an address match the primary street name of a GIS feature to be considered valid. Because the civic location does not match the address point's primary street name, the LVF claims the address is invalid. LVF B, however, does match against street name aliases when validating. It finds a match on the street name alias of COUNTY ROAD 30, and thus claims the address is valid. Both LVFs produce different baseline validation results because their use of street name aliases varies.

A.5 Differences in GIS Field Mappings

Section 2.4.1.5 details a problem that can lead to LVF inconsistency, if LVFs have inconsistent configuration settings that specify which GIS data attribute fields represent address elements found in PIDF-LO. This example illustrates the possible differing behavior of two LVFs. Given this civic location and a site/structure address point in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <country>US</country>
  <A1>MN</A1>
  <A2>HENNEPIN COUNTY</A2>
  <A3>MAPLE GROVE</A3>
  <RD>93RD</RD>
  <STS>AVENUE</STS>
  <POD>NORTH</POD>
  <HNO>11000</HNO>
  <PC>55369</PC>
  <PCN>OSSEO</PCN>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2</valid>
  <invalid>ca:A3</invalid>
  <unchecked>ca:RD ca:STS ca:POD ca:HNO ca:PCN ca:PC</unchecked>
</locationValidation>
```

LVF B might return the following, different result:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:A3 ca:RD ca:STS ca:POD ca:HNO ca:PCN ca:PC</valid>
  <invalid></invalid>
  <unchecked></unchecked>
</locationValidation>
```

A summary of the proffered civic address, its matching road centerline, and the results from LVF A and LVF B are as follows, listed using PIDF-LO element ordering:

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country	US	country>US</country>	Valid	Valid
A1	State	MN	<A1>MN</A1>	Valid	Valid
A2	County	HENNEPIN COUNTY	<A2>HENNEPIN COUNTY</A2>	Valid	Valid
A3	Inc_Muni	MAPLE GROVE	<A3>MAPLE GROVE</A3>	Invalid	Valid

RD	StreetName	93RD	<RD>93RD</RD>	Unchecked	Valid
STS	St_PosTyp	AVENUE	<STS>AVENUE</STS>	Unchecked	Valid
POD	St_PosDir	NORTH	<POD>NORTH</POD>	Unchecked	Valid
	MSAGComm	MSAGVILLE			
HNO	Add_Number	11000	<HNO>11000</HNO>	Unchecked	Valid
PC	Post_Code	55369	<PC>55369</PC>	Unchecked	Valid
PCN	Post_Comm	OSSEO	<PCN>OSSEO</PCN>	Unchecked	Valid

LVF A claims the address is invalid, because it was configured by its administrator to match A3 values against the address point layer's MSAG Community attribute. LVF B returns a different baseline result, however; it claims the address is valid. Unlike LVF A, the administrator of LVF B configured it to match A3 values against the address point's Incorporated Municipality attribute.

A.6 Analyzing Address Elements in a Different Order

Section 2.4.2.1 details a problem that can lead to LVFs returning valid/invalid/unchecked elements in varying order, if LVFs do not follow the hierarchical approach to validating addresses as defined in section 2.5.3. This example illustrates the possible differing behavior of two LVFs. Given the following civic location and site/structure address point in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr">
  <country>US</country>
  <A1>TX</A1>
  <A2>WALLER COUNTY</A2>
  <A3>MAGNOLIA</A3>
  <RD> WOODWA</RD>
  <STS> DRIVE </STS>
  <HNO>137</HNO>
  <PC>77355</PC>
  <PCN>MAGNOLIA</PCN>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:A3</valid>
  <invalid>ca:RD</invalid>
  <unchecked>ca:STS ca:HNO ca:PCN ca:PC</unchecked>
</locationValidation>
```

LVF B might return the following, different result:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pdf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A2 ca:A3</valid>
  <invalid>ca:RD</invalid>
```

```
<unchecked>ca:HNO ca:STS ca:PCN ca:PC</unchecked>
</locationValidation>
```

A summary of the proffered civic address, its matching road centerline, and the results from LVF A and LVF B are as follows, listed using PIDF-LO element ordering. Note the indication of the order of the Unchecked elements.

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country	US	<country>US</country>	Valid	Valid
A1	State	TX	<A1>TX</A1>	Valid	Valid
A2	County	WALLER COUNTY	<A2>WALLER COUNTY</A2>	Valid	Valid
A3	Inc_Muni	MAGNOLIA	<A3>MAGNOLIA</A3>	Valid	Valid
RD	StreetName	WOODWAY	<RD> WOODWA</RD>	Invalid	Invalid
STS	St_PosTyp	DRIVE	<STS>DRIVE</STS>	Unchecked (1)	Unchecked (2)
HNO	Add_Number	137	<HNO>137</HNO>	Unchecked (2)	Unchecked (1)
PC	Post_Code	77355	<PC>77355</PC>	Unchecked (4)	Unchecked (4)
PCN	Post_Comm	MAGNOLIA	<PCN>MAGNOLIA</PCN>	Unchecked (3)	Unchecked (3)

LVF A and LVF B both claim the location is invalid; they both provide consistent *baseline* results. However, their results are inconsistent because the order of the unchecked elements varies. LVF A follows the hierarchical approach to validating addresses as defined in section 2.5.3, and lists STS before HNO. LVF B, however, does not follow the hierarchical approach and lists HNO before STS. As such, the LVFs provide inconsistent results.

A.7 Claiming Different Elements are Invalid

Section 2.4.2.2 details a problem that can lead to LVFs returning different valid/invalid/unchecked elements, if LVFs do not follow the same general guidelines regarding how many invalid elements are returned when validation fails, such as those described in section 2.5.3. This example illustrates the possible differing behavior of two LVFs. Given the following civic location and site/structure address point in the LVF:

```
<civicAddress xmlns="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <country>US</country>
  <A1>TX</A1>
  <A2>NONEXISTENT COUNTY</A2>
  <A3>MAGNOLIA</A3>
  <RD> NONEXISTENT </RD>
  <STS>STREET</STS>
  <HNO>137</HNO>
```

```
<PC>77355</PC>
<PCN>MAGNOLIA</PCN>
</civicAddress>
```

LVF A might return the following result (an excerpt of its LoST findServiceResponse):

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1</valid>
  <invalid>ca:A2 </invalid>
  <unchecked>ca:A3 ca:RD ca:STS ca:HNO ca:PCN ca:PC </unchecked>
</locationValidation>
```

LVF B might return the following, different result:

```
<locationValidation xmlns:ca="urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr">
  <valid>ca:country ca:A1 ca:A3 ca:HNO ca:PCN ca:PC </valid>
  <invalid>ca:A2 ca:RD ca:STS</invalid>
  <unchecked></unchecked>
</locationValidation>
```

A summary of the proffered civic address, its matching road centerline, and the results from LVF A and LVF B are as follows, listed using PIDF-LO element ordering:

PIDF-LO Element Name	GIS Attribute Name	GIS Attribute Value	Proffered Civic Location Value	LVF A Validation Result	LVF B Validation Result
country	Country	US	<country>US</country>	Valid	Valid
A1	State	TX	<A1>TX</A1>	Valid	Valid
A2	County	WALLER COUNTY	<A2>NONEXISTENT COUNTY</A2>	Invalid	Invalid
A3	Inc_Muni	MAGNOLIA	<A3>MAGNOLIA</A3>	Unchecked	Valid
RD	StreetName	WOODWAY	<RD> NONEXISTENT </RD>	Unchecked	Invalid
STS	St_PosTyp	DRIVE	<STS>STREET</STS>	Unchecked	Invalid
HNO	Add_Number	137	<HNO>137</HNO>	Unchecked	Valid
PC	Post_Code	77355	<PC>77355</PC>	Unchecked	Valid
PCN	Post_Comm	MAGNOLIA	<PCN>MAGNOLIA</PCN>	Unchecked	Valid

LVF A and LVF B both claim the location is invalid; they both provide consistent *baseline* results. However, their results are inconsistent from the perspective of the civic location elements they claim are valid, invalid, and those that were not checked during validation. The contents of their valid/invalid/unchecked element lists differ, so they, too, provide inconsistent validation results.

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